Wet Beriberi and Thiamine Deficiency

A red, burning tongue and peripheral edema have also been observed with wet beriberi.

Laboratory Testing

Diagnosis of thiamine deficiency is made by measurement of erythrocyte thiamine transketolase activity or blood thiamine concentration. The more reliable measure is erythrocyte thiamine transketolase activity before and after thiamine pyrophosphate stimulation, expressed as the percentage of thiamine pyrophosphate effect. Normal values are up to 15 percent.

Treatment

Thiamine is a cofactor in multiple metabolic pathways, so daily requirements are based on caloric intake, with current recommendations of 0.5 mg per 1000 kcal.

Treatment can be administered intravenously, intramuscularly, or orally. For beriberi, treatment is typically initiated with intravenous or intramuscular thiamine at 50 to 100 mg per day for 7 to 14 days, followed by oral supplementation until full recovery is achieved.

Vitamin B_2 (Riboflavin)

Etiology and Pathogenesis

Riboflavin was first identified in 1879 as a fluorescent yellow-green compound in milk. It functions in two coenzymes: flavin mononucleotide (FMN) and flavin-adenine dinucleotide (FAD), which play key roles in oxidation-reduction reactions during cellular respiration and oxidative phosphorylation. These coenzymes are also involved in vitamin B_6 (pyridoxine) metabolism.

Recent studies suggest riboflavin deficiency may contribute to elevated plasma homocysteine levels, impaired iron metabolism, and night blindness.

Dietary sources include dairy products, nuts, meat, eggs, whole grains, enriched breads, fatty fish, and green leafy vegetables. Most dietary riboflavin is present as FAD or FMN, which are hydrolyzed to free riboflavin by brush border enzymes in the small intestine. Free riboflavin is then absorbed via active transport in the proximal small bowel.

Deficiency can result from:

- Inadequate intake
- Malabsorption
- Phototherapy

At-risk populations include alcoholics, the elderly, and adolescents due to poor nutrition. Riboflavin deficiency is endemic in parts of India, China, and Iran where diets rely heavily on unenriched cereals.

Infants of riboflavin-deficient mothers are at risk because breast milk levels reflect maternal status. After weaning, deficiency risk increases if alternative sources like milk are not introduced. Protein-energy malnutrition (PEM) exacerbates deficiency by impairing the kidney's ability to conserve riboflavin.

Phototherapy for neonatal jaundice causes photodegradation of riboflavin. Certain drugs also interfere with riboflavin metabolism:

- **Chlorpromazine** and tricyclic antidepressants inhibit gastrointestinal absorption.
- **Borate** displaces riboflavin from binding sites, increases urinary excretion, and inhibits riboflavin-dependent enzymes.

Clinical Findings

Acute riboflavin deficiency presents with deep red erythema, epidermal necrolysis, and mucositis. Symptom severity correlates with the degree of deficiency.

Chronic deficiency, or **ariboflavinosis**, develops 3 to 5 months after inadequate intake. Skin and mucous membrane changes are predominant.

- **Angular stomatitis**: Begins as small papules at the mouth corners, progressing to ulcers and macerated, bleeding fissures that extend laterally.
- **Cheilosis**: Marked by erythema, dryness (xerosis), and vertical fissuring of the lips.
- **Glossitis**: Early stage shows prominent lingual papillae; as deficiency progresses, papillae atrophy, resulting in a smooth, swollen, magenta-colored tongue.